

Class-Resolved Cross-City Transfer of YOLO Vehicle Detectors: A Dhaka-to-Chattogram Study

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Abstract—Object detectors trained on one Bangladeshi city’s road imagery are known to lose accuracy when deployed in another, and detectors trained on structured traffic are known to transfer poorly to unstructured South Asian traffic. What is not established is whether a same-city, same-protocol transfer gap is spread evenly across vehicle types, or concentrated in specific ones. Holding annotation protocol and camera viewpoint fixed, we train YOLOv8s detectors on Dhaka dash-camera imagery from the BadODD dataset and evaluate them on a held-out Chattogram set, against a Chattogram-trained reference, using a 3-fold temporal-block cross-validation design plus two leave-one-corridor-out robustness folds. We find the transfer gap is not uniform: it is smallest for globally-standard motorised classes (car, bus, truck, motorbike; AP50 gap 0.061) and 3–4× larger for two locally-specific informal and non-motorised modes – CNG/battery auto-rickshaws (AP50 gap 0.197) and cycle-rickshaws (AP50 gap 0.225) – with an even larger gap in recall, the operating point that determines whether a vehicle is counted at all. The pattern replicates in both robustness folds and under a second, newer detector architecture (YOLO26s). Because vehicle counts derived from such detectors feed downstream traffic-signal timing, this class-concentrated blind spot has a direct equity implication we discuss.

Index Terms—object detection, domain transfer, heterogeneous traffic, Bangladesh, YOLO

I. INTRODUCTION

Urban traffic in Bangladeshi cities is a heterogeneous, largely non-lane-based mix, with informal and non-motorised vehicles sharing the carriageway alongside cars, buses and trucks – a mix that differs sharply from the structured traffic most detection benchmarks are built on. Large driving-scene benchmarks such as BDD100K [1] and the original COCO object-detection benchmark [2] – the source of the AP50 metric and evaluation protocol we use throughout this paper – are drawn overwhelmingly from structured, lane-based traffic and do not include informal-transport classes at all. Object detectors intended to feed automated traffic monitoring or signal control must therefore work well on exactly the vehicle types structured-traffic benchmarks under-represent.

Two facts are already established in the literature and are not this paper’s contribution. First, detectors trained on structured traffic transfer poorly to unstructured, developing-city traffic: fixed-camera evaluation on BMD-45 reports a 2.5× mAP gap between a structured-traffic-trained detector and an in-domain one on an expert-annotated reference set [3]; the gap is smaller but still consistently positive ($\approx 1.1\times$ mAP@0.5:0.95) in that paper’s main architecture-matched cross-dataset benchmark table for a comparably-sized YOLO variant, so the size of

a structured-to-unstructured gap is itself evaluation-protocol-dependent. Second, cross-city detection generalisation is an active, separately benchmarked problem, evidenced by ongoing challenge tracks [4] and multi-task cross-city benchmarks such as CityTransfer-Bench [5]. Neither source reports whether a same-city, same-annotation-protocol transfer gap concentrates on specific vehicle classes.

This paper isolates that question. Using BadODD [6], a single dashcam dataset annotated under one protocol across nine Bangladeshi districts including both Dhaka and Chattogram, we hold camera viewpoint and annotation protocol fixed and ask whether a detector trained on Dhaka imagery, once evaluated on Chattogram imagery, shows a performance drop that is uniform across vehicle classes or instead concentrated in specific modes – particularly the informal and non-motorised ones that existing structured-traffic benchmarks do not cover at all? We report recall (whether a vehicle is detected at all) as the primary quantity of interest, since a missed vehicle is what would later bias a classified count in a downstream traffic-monitoring pipeline.

Our contribution is a controlled, class-resolved measurement of this gap, not a new detector or dataset. We find the gap is 3–4× larger for two locally-specific modes – CNG/battery auto-rickshaws and cycle-rickshaws – than for globally-standard motorised classes, and that this pattern is robust both to the cross-validation design used to measure it and to the choice of detector architecture.

II. RELATED WORK

BadODD [6] is the dataset this paper builds on; its class taxonomy, defined by power source and size rather than local vehicle name, is what makes a same-protocol Dhaka-Chattogram comparison possible in the first place (Section III).

BMD-45 [3] establishes that a structured-to-unstructured camera domain gap exists, attributing it to viewpoint and elevation differences between structured and CCTV-style fixed cameras. That is a different axis from the one studied here: BadODD is dash-mounted in both cities, so our comparison holds viewpoint fixed and isolates city/location effects instead. BMD-45’s benchmark tables do not report a class-wise breakdown of its transfer gap, so whether that gap is itself class-concentrated is, to our knowledge, still open. Other South Asian unstructured-traffic datasets such as IDD [7] cover a broader mix of road users but are single-region collections and do not offer the same-protocol multi-city split that lets

us hold annotation protocol fixed while varying city, the way BadODD does.

Bangladeshi-vehicle YOLO benchmarks, e.g. [8], confirm YOLO variants have already been evaluated extensively on Bangladeshi road imagery and that rare classes collapse under class imbalance – consistent with what we observe for under-30-instance classes in Section IV. Rare-class collapse under long-tailed label distributions is itself a well-studied general problem in object detection [9], [10], but that line of work targets a fixed in-domain class imbalance; we study a case where class imbalance interacts with a cross-city domain shift, and neither [8] nor the general long-tail literature performs a cross-city or cross-domain transfer evaluation.

Detection bias against vulnerable road users is a recognised problem in the wider perception literature [11], [12], studied there on structured-traffic data (nuScenes, BDD100K) and framed around demographic or safety disparities among pedestrians. Our framing differs: the downstream harm we are concerned with is a counting/allocation one (Section V), the disparity is across vehicle *class* rather than pedestrian demographic group, and the setting is unstructured, non-lane-based traffic rather than structured urban driving.

CityTransfer-Bench [5] benchmarks cross-city generalisation broadly across perception, segmentation and planning, and proposes closing the gap via synthetic target-city imagery. It is not class-resolved and does not target Bangladeshi informal transport modes specifically.

More broadly, the Dhaka-to-Chattogram gap we measure is an instance of the domain-shift problem surveyed in [13]: a model trained on a source distribution (Dhaka) underperforming on a related but distinct target distribution (Chattogram) without target-domain adaptation. We do not apply a domain-adaptation method here – our contribution is measuring where, class-wise, that gap is largest – but the survey’s taxonomy of feature-level vs. instance-level adaptation is a natural starting point for closing the gap this paper documents; concrete instance-level methods such as [14] already adapt detectors across exactly this kind of camera/viewpoint domain shift, but have not, to our knowledge, been applied to a same-city class imbalance of the kind reported here.

III. DATASET AND EXPERIMENTAL SETUP

A. Dataset

We use the Zenodo release of BadODD (DOI 10.5281/zenodo.13823687, CC BY 4.0), 10,032 fully labelled dash-camera images across 9 Bangladeshi districts, 11 vehicle classes with non-zero Chattogram or Dhaka instances (`train` and `wheelchair` have zero Chattogram instances and are dropped). Classes are defined by physical characteristic rather than local name – notably, `three_wheeler` (paddle-powered, i.e. the cycle-rickshaw) is a distinct class from `auto_rickshaw` (gas/electric-powered, covering both CNG auto-rickshaws and battery-powered “tomtom”/easy-bikes). Of the 11 trained classes, 7 have at least 30 Chattogram test instances and are reported individually (Table I); the remaining 4

(`bicycle`, `cart_vehicle`, `priority_vehicle`, `construction_vehicle`) are below that threshold and are reported only as part of mode-group aggregates, weighted by instance count so they cannot dominate a group’s number.

We group the 7 headline classes into 4 mode groups: *pedestrian* (person), *non-motorised* (three-wheeler/cycle-rickshaw, plus the below-threshold bicycle and cart vehicle), *informal motorised* (auto-rickshaw), and *formal motorised* (car, bus, truck, motorbike, plus the below-threshold priority and construction vehicles).

B. Splits and cross-validation

Chattogram’s 3 continuous video drives are each split into 3 temporal blocks with a ≥ 15 -second buffer between blocks to avoid near-duplicate leakage, giving a 3-fold cross-validation design. Dhaka training sets are subsampled (seed 42) to match each Chattogram fold’s training-set size and day/night ratio, so that a Dhaka-vs-Chattogram comparison is not confounded by training-set size. Two further robustness folds hold out one entire daytime corridor each (no temporal-block cuts), subsampled to the same training size, to check the main result is not an artifact of the temporal-block partitioning. Full split counts are in `notes/splits.md`.

C. Models and training

YOLOv8s (Ultralytics 8.4.155), a single-stage detector in the YOLO lineage that began with the original YOLO/YOLOv3 formulation [15] and has been reviewed in detail up to YOLOv8 by [16], COCO-pretrained, 640px input, fixed hyperparameters and seed (42, deterministic) across every run, 100 epochs, no early stopping, no model-selection on evaluation data (`last.pt` used throughout). Two experiment families: E1 (Dhaka-trained, 3 folds) and E2 (Chattogram-trained, 3 folds), plus the 2 robustness-fold models. Every run exports raw predictions at confidence 0.001 in COCO format, so all metrics below are recomputed offline. All 8 runs are then repeated with YOLO26s [17] (same Ultralytics version, COCO-pretrained), with splits, seed and hyperparameters unchanged – including the SGD optimizer, rather than YOLO26’s default MuSGD – so that only the model differs, to test whether the result depends on the architecture. Code, the offline metric-recomputation pipeline and every prediction file behind the numbers below are available at <https://anonymous.4open.science/r/ecce2027-4E83>.

D. Metrics

Per-class AP50 (COCO-style, 101-point interpolation, following the original COCO evaluation protocol [2]) and recall at a fixed operating point, chosen per model by maximising pooled F1 over the 7 headline classes on that model’s own in-domain evaluation fold and held fixed for every other evaluation of that model. 95% confidence intervals are image-level bootstrap (resampling images with replacement; local implementation validated against Kaggle’s own pycocotools output, exact match on the non-bootstrapped point estimate).

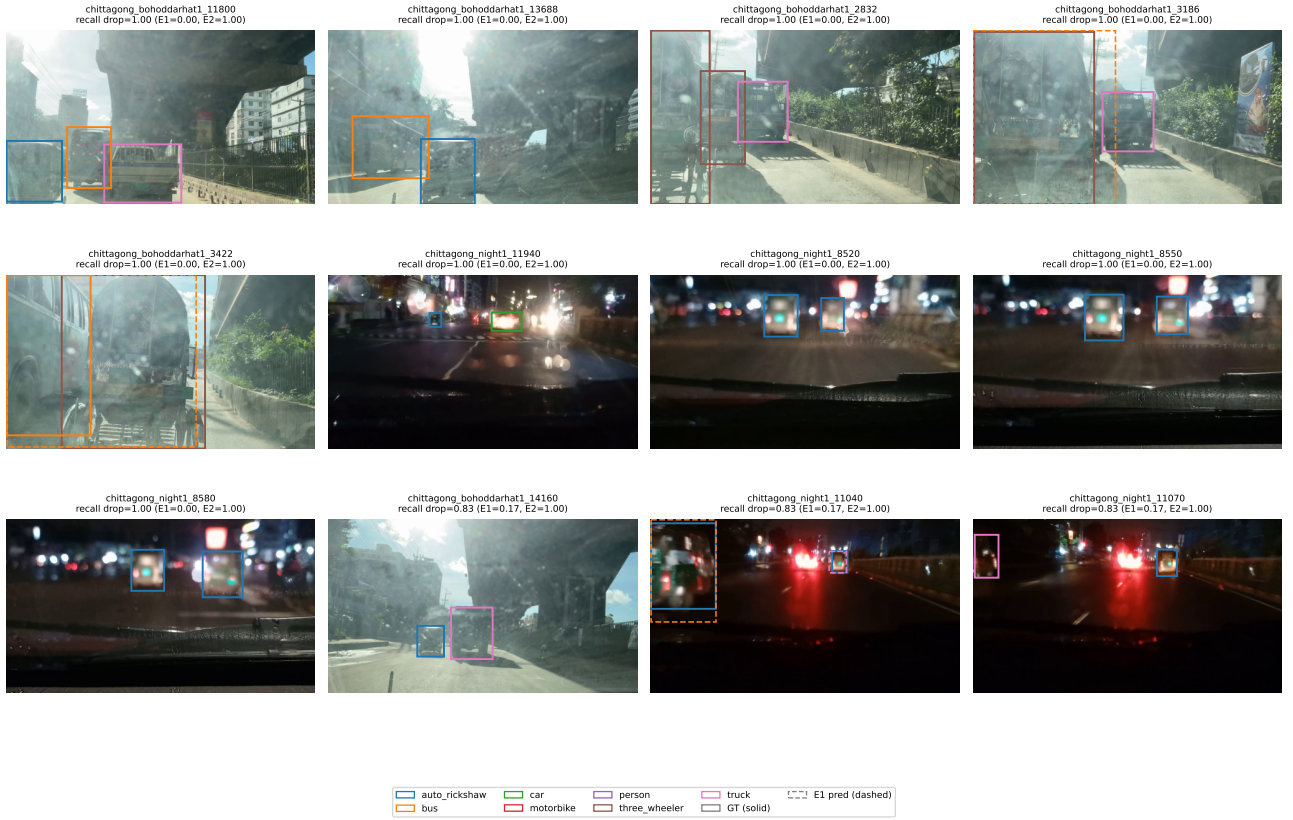


Fig. 1. The 12 `ctg_pooled_eval` images with the largest per-image recall drop (E2 recall – E1 recall, headline classes, ≥ 2 GT boxes), selected automatically by `src/run_wp4_qualitative.py` – not hand-picked. Solid boxes: ground truth. Dashed boxes: E1 (`e1_0`) detections above its operating threshold. Most panels show E1 missing every GT box in the frame. Discussed alongside the quantitative results in Section IV.

TABLE I

DHAKA-TRAINED (E1, MEAN OF 3 FOLDS) VS. CHATTOGRAM-TRAINED REFERENCE (E2, OUT-OF-FOLD POOLED), EVALUATED ON THE IDENTICAL 1,121-IMAGE CHATTOGRAM TEST SET. AP50 GAP = E2 – E1, WITH 95% IMAGE-LEVEL BOOTSTRAP CI (300 RESAMPLES).

Class	n	E1 AP50	E2 AP50	AP50 gap [95% CI]
Auto-rickshaw	1796	0.558	0.755	+0.197 [+0.180, +0.216]
Bus	411	0.308	0.325	+0.017 [-0.035, +0.070]
Car	1122	0.521	0.614	+0.094 [+0.069, +0.117]
Motorbike	401	0.414	0.369	-0.045 [-0.096, +0.002]
Person	2509	0.272	0.366	+0.094 [+0.077, +0.111]
Three-wheeler (cycle-rick.)	988	0.327	0.561	+0.234 [+0.203, +0.267]
Truck	668	0.295	0.393	+0.098 [+0.056, +0.137]

IV. RESULTS

Table I compares E1 (mean of the 3 Dhaka-trained folds) against E2 (the 3 Chattogram-trained folds’ out-of-fold pooled predictions), both evaluated on the same 1,121-image Chattogram test set. Every headline class except `motorbike` shows a positive AP50 gap (E2 above E1), but the size of that gap varies by more than an order of magnitude across classes.

Table II aggregates by mode group, weighted by each class’s

TABLE II

TRANSFER GAP AGGREGATED BY MODE GROUP, WEIGHTED BY EACH CLASS’S CHATTOGRAM INSTANCE COUNT.

Mode group	n	E1 AP50	E2 AP50	AP50 gap
Formal motorised	2621	0.410	0.471	+0.061
Pedestrian	2509	0.272	0.366	+0.094
Informal motorised	1796	0.558	0.755	+0.197
Non-motorised	1026	0.315	0.540	+0.225

Chattogram instance count. The formal-motorised group – car, bus, truck, motorbike, the classes any general-purpose, COCO-trained detector would already recognise – shows the smallest gap (0.061 AP50, 0.081 recall). The two locally-specific groups show substantially larger gaps: informal motorised (auto-rickshaw) at 0.197 AP50 / 0.263 recall, and non-motorised (cycle-rickshaw-dominated) at 0.225 AP50 / 0.277 recall – both roughly 3–4 $\times$ the formal-motorised gap.

Per-class 95% bootstrap CIs (Table I) show this is not uniformly noise: the car, person, truck, three_wheeler and auto_rickshaw gaps all exclude zero, and the two largest gaps – the two locally-specific classes, auto-rickshaw

TABLE III
MODE-GROUP AP50 AND RECALL TRANSFER GAPS WITH 95%
BOOTSTRAP CI (300 SHARED IMAGE RESAMPLES ACROSS ALL 11
CLASSES).

Mode group	AP50 gap [95% CI]	Recall gap [95% CI]
Formal motorised	+0.061 [+0.043, +0.081]	+0.081 [+0.064, +0.100]
Pedestrian	+0.094 [+0.073, +0.112]	+0.119 [+0.100, +0.135]
Informal motorised	+0.197 [+0.180, +0.217]	+0.263 [+0.241, +0.282]
Non-motorised	+0.225 [+0.195, +0.256]	+0.277 [+0.245, +0.305]

TABLE IV
OUR HEADLINE-CLASS MEAN AP50 AGAINST DETECTOR PERFORMANCE
REPORTED BY PRIOR WORK ON RELATED BANGLADESHI/SOUTH ASIAN
VEHICLE-DETECTION TASKS. TASKS, CLASS SETS AND TRAINING
BUDGETS DIFFER ACROSS ROWS (SEE SECTION IV); THESE ARE CONTEXT
POINTS, NOT A CONTROLLED COMPARISON.

Source	Task	Detector	Metric
This work (E2, in-domain)	BadODD, Ctg., 7-class	YOLOv8s	0.483 mAP50
This work (E1, Dhaka→Ctg.)	BadODD, Ctg., 7-class	YOLOv8s	0.385 mAP50
BadODD baseline [6]	BadODD, all-district	YOLOv8	0.70 mAP ^a
Hossain et al. [8]	BD urban, 29-class	YOLOv8m	0.625 mAP50
BMD-45 [3]	UA-DETRAC→BMD-45 ^b	YOLOv12-S	0.38→0.42 mAP50:95

^aMetric unspecified as mAP50 vs. mAP50:95 in the source; 10-epoch baseline, not a tuned reference. ^bCross-dataset transfer, not cross-city; architecture-matched row from the source’s main benchmark table, shown for scale rather than direct comparison.

[0.180, 0.216] and three-wheeler – are among the most clearly significant. Following the stated protocol for a CI that spans zero, we report *bus* ($[-0.035, +0.070]$) and *motorbike* ($[-0.096, +0.002]$) as *no evidence of a class-level difference* at this sample size, not as null findings for the formal-motorised group as a whole – the group’s weighted point estimate is still driven upward by *car* and *truck*, both individually significant.

Table III gives the mode-group gaps their own 95% bootstrap CI (300 resamples, shared across all 11 classes per resample so the group aggregate reflects one coherent resample of the underlying images rather than 11 independent ones). All four groups’ CIs exclude zero, so every group has *some* real gap at this sample size – but the formal-motorised CI, [+0.043, +0.081], sits roughly 3–4× lower than the two locally-specific groups’ intervals and overlaps neither of them (it overlaps the pedestrian interval only marginally). The claim is therefore not “formal motorised has no gap,” but that its gap is consistently and substantially smaller than the two locally-specific groups’, with non-overlapping confidence intervals.

One class runs against the pattern and is reported straight, not hidden: *motorbike* shows a small *negative* AP50 gap (E1 above E2 by 0.045), i.e. the Dhaka-trained model slightly outperforms the Chattogram-trained reference on motorbikes in Chattogram. *bus* recall is also flat-to-negative. So the formal-motorised group is not uniformly gap-free – it is consistently smaller and sometimes reversed, while the two locally-specific groups are consistently and substantially worse.

A. Comparison with prior benchmarks

Table IV places our headline-class mean AP50 (0.483 for E2, in-domain; 0.385 for E1, cross-city) alongside numbers reported by the three Bangladeshi/South Asian vehicle-detection papers already discussed in Section II. These are not a controlled comparison – class sets, training budgets and evaluation protocols all differ – but they give useful scale. Our E2 number sits below Hossain et al.’s YOLOv8m result (0.625 mAP50) on a 29-class, non-transfer Bangladeshi dataset [8] and below BadODD’s own 10-epoch baseline (0.70, metric unspecified) [6]; both prior numbers average over an easier, mostly formal-motorised class mix and neither reports a per-class or cross-city breakdown, so a lower pooled number here is consistent with, not contrary to, our finding that two locally-specific classes drag the pooled mean down substantially (Table I). The BMD-45 comparison [3] is included for scale only: it measures a cross-dataset (UA-DETRAC→BMD-45) transfer gap for a similarly-sized YOLO variant, not a same-dataset cross-city gap, and its own architecture-matched benchmark table shows a considerably smaller gap ($\approx 1.1\times$ mAP50:95) than the 2.5x figure quoted from its expert-reference-set evaluation (Section I).

B. Robustness check

Both leave-one-corridor-out folds (rob_a, rob_b), trained and evaluated without any temporal-block cuts, reproduce the same ranking: the three-wheeler (cycle-rickshaw) AP50 gap against the mean of the 3 Dhaka-trained models is +0.278 and +0.162 in the two corridors respectively, larger than every formal-motorised class’s gap in either corridor (results/per_class_ap.csv). This indicates the main result is not an artifact of the temporal-block cross-validation design.

C. Architecture robustness check

To check whether this pattern is specific to YOLOv8s or holds more generally, we repeat the full E1/E2 experiment – identical splits, seed and hyperparameters, only the model swapped – with YOLO26s, Ultralytics’ current flagship detector [17]. The same ordering holds: the formal-motorised gap is again smallest (0.054 AP50), while the informal-motorised (0.185) and non-motorised (0.227) gaps are 3–4× larger, closely matching the YOLOv8s ratios (Table II). YOLO26s is the stronger detector – headline-class mean AP50 rises from 0.385 to 0.426 for E1 and from 0.483 to 0.523 for E2 – but both curves rise together, so the better architecture does not close the gap between them. The leave-one-corridor-out robustness folds replicate as well: under YOLO26s the three-wheeler AP50 gap is +0.205 and +0.157 in the two corridors (cf. +0.278/+0.162 for YOLOv8s), again larger than every formal-motorised class’s gap in the same corridor. The class-concentrated transfer gap is therefore not an artifact of the YOLOv8s architecture specifically.

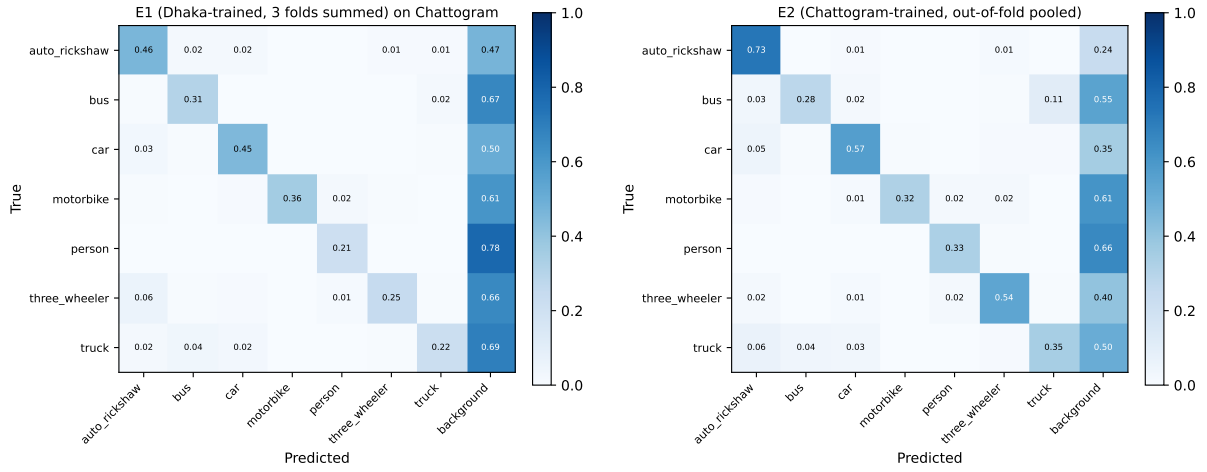


Fig. 2. Row-normalised confusion matrices (headline classes + background; IoU ≥ 0.5 greedy match) for E1 (left, 3 Dhaka folds summed) and E2 (right, Chattogram out-of-fold pooled), both on `ctg_pooled_eval`. A GT box with no matching prediction counts as “background” in that row (a miss); a prediction with no matching GT counts as “background” in that column (a false positive).

TABLE V

AP50 TRANSFER GAP BY MODE GROUP, STRATIFIED BY DAY VS. NIGHT. RESTRICTED TO HEADLINE CLASSES ONLY (BELOW-THRESHOLD CLASSES NOT INCLUDED: NON-MOTORISED = `THREE_WHEELER` ONLY HERE, VS. + `BICYCLE/CART` IN TABLE 2).

Mode group	Day AP50 gap	Night AP50 gap
Formal motorised	+0.125	+0.017
Pedestrian	+0.109	+0.068
Informal motorised	+0.144	+0.219
Non-motorised	+0.253	+0.136

D. Confusion patterns

Figure 2 breaks the recall gap down into misses vs. misclassifications. For E1, the background column (missed detections) is the dominant off-diagonal entry for every headline class, and is visibly larger for `auto_rickshaw` and `three_wheeler` than for `car/person` – consistent with the recall-gap ranking in Table I. The largest specific misclassification in the E1 matrix is `three_wheeler` predicted as `auto_rickshaw` under E1 (6% of true `three_wheeler` instances), plausibly reflecting genuine visual similarity between cycle-rickshaws and motorised three-wheelers rather than a purely random error – this confusion is smaller under E2, suggesting Chattogram-specific visual cues (not present in Dhaka training data) help separate the two classes.

E. Day/night stratification

Table V splits the mode-group AP50 gap by day (456 images) vs. night (665 images), the same day/night proportions reported in `notes/splits.md`. The informal-motorised and non-motorised gaps stay large and positive in both conditions. The formal-motorised gap, however, is not uniformly small: by day it is +0.125, close to the informal-motorised gap and driven mainly by `truck` and `bus`, while at night it nearly vanishes (+0.017) because `bus`, `motorbike` and `truck`

individually reverse sign (E1 outperforming E2). The pooled formal-motorised number reported earlier therefore averages over two quite different regimes, and the class concentration of the gap is sharpest at night – a day/night interaction worth flagging rather than averaging away.

F. Qualitative failures

Figure 1 (Section III) shows the images E1 fails on most severely relative to E2. Several are low-light, glare-affected frames where E1 detects nothing at all while E2 recovers every box – consistent with the day/night finding above, though the auto-selection was not conditioned on time of day. Because selection is by recall drop alone, several frames come from the same short sequence (`chittagong_night1`, frames 8520–8580); a de-duplicated selection (e.g. one frame per sequence) is a natural refinement for a later revision.

G. Reverse transfer

Evaluating the Chattogram-trained models on their corresponding Dhaka fold (a different direction of transfer) shows every class degrading relative to that fold’s own Dhaka-trained in-domain score, as expected. Here the classes hit hardest are `truck` and `bus` (AP50 dropping by 0.24–0.39), not the informal or non-motorised classes – so the class-concentration of the transfer penalty is direction-dependent, not a fixed per-class difficulty ranking (Section V).

V. DISCUSSION

A. Why this matters beyond detection accuracy

A detector deployed for automated traffic counting does not report AP50; it reports a count. A class with lower recall is one whose vehicles are disproportionately missing from that count. If informal and non-motorised modes are the ones most often missed when a detector is transferred to a new city without local retraining, any downstream use of those

counts – e.g. converting classified counts to passenger-car-unit demand for signal-timing allocation – would systematically under-represent the travel demand of the road users least able to absorb the resulting delay. This concern is not hypothetical: Dhaka-specific traffic-signal control work already proposes using detected vehicle counts directly for congestion-optimised signal timing [18], which would make the detection-level blind spot documented here a direct input to such a pipeline rather than a purely abstract downstream risk. This paper measures the detection-level premise of that chain; we do not measure the downstream counting or allocation effect here.

B. Limitations

The camera viewpoint in both cities is a dash-mounted smartphone, not a fixed traffic camera of the kind an actual deployed monitoring system would use – a separate, already-established axis of domain gap [3], orthogonal to the one studied here. All data comes from one dataset and one annotation protocol; class definitions are the dataset authors', not ours. City is not the only variable that differs between the Dhaka and Chattogram splits – lighting and road-type mix also differ. We match day/night ratio and training-set size by construction between the Dhaka and Chattogram training sets (Section III), and stratify the headline result by day/night directly (Table V) rather than only controlling for it in aggregate; we do not control for road-type mix. The 4 below-threshold classes cannot support an individual claim at all; they are reported only in aggregate. Both architectures tested are single-stage Ultralytics YOLO models; two-stage and transformer-based detectors are untested. The qualitative failure selection (Figure 1) is unconditioned on sequence identity, so several selected frames come from the same short video clip rather than being independent samples.

VI. CONCLUSION

Holding annotation protocol and camera viewpoint fixed, we find that the Dhaka-to-Chattogram detection transfer gap is not spread evenly across vehicle classes. It is smallest for car, bus, truck and motorbike – classes any general-purpose detector would already know – and 3–4× larger for two locally-specific informal and non-motorised modes, auto-rickshaws and cycle-rickshaws, with an even larger gap in recall specifically. The pattern replicates under an independent leave-one-corridor-out robustness check and under a newer detector architecture (YOLO26s), which raises absolute accuracy but leaves the gap essentially unchanged. Because missed vehicles are what would later bias a classified traffic count, this class-concentrated blind spot is a plausible mechanism for downstream inequity in any traffic-management system built on a detector transferred without local retraining – a link we intend to measure directly in future work.

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